SG No. 47 D_{2h}^{1} $Pmmm$ [orthorhombic]

* plus set: $+[0, 0, 0]$
* Wyckoff site: `1a`, site symmetry: `mmm`

Table 1: Wyckoff bond: `1a@1a`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, 0, Z]$	[0, 0, 0]	`[1,2,-3,-4,-5,-6,7,8]`

Table 2: Wyckoff bond: `1b@1a`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, Y, 0]$	[0, 0, 0]	`[1,-2,3,-4,-5,6,-7,8]`

Table 3: Wyckoff bond: `1c@1a`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[X, 0, 0]$	[0, 0, 0]	`[1,-2,-3,4,-5,6,7,-8]`

Table 4: Wyckoff bond: `2d@1a`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, Y, 0]$	[0, 0, 0]	`[1,-2,-5,6]`
2	$[-X, Y, 0]$	[0, 0, 0]	`[3,-4,-7,8]`

Table 5: Wyckoff bond: `2e@1a`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	[0, 0, 0]	`[1,-3,-5,7]`
2	$[-X, 0, Z]$	[0, 0, 0]	`[2,-4,-6,8]`

Table 6: Wyckoff bond: `2f@1a`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	[0, 0, 0]	`[1,-4,-5,8]`
2	$[0, -Y, Z]$	[0, 0, 0]	`[2,-3,-6,7]`

Table 7: Wyckoff bond: `4g@1a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	[0, 0, 0]	`[1,-5]`
2	$[-X, -Y, Z]$	[0, 0, 0]	`[2,-6]`
3	$[-X, Y, -Z]$	[0, 0, 0]	`[3,-7]`

continued ...

Table 7

No.	vector	center	mapping
4	$[X, -Y, -Z]$	$[0, 0, 0]$	`[4,-8]`

* Wyckoff site: `1b`, site symmetry: `mmm`

Table 8: Wyckoff bond: `1a@1b`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,2,-3,-4,-5,-6,7,8]`

Table 9: Wyckoff bond: `1b@1b`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-2,3,-4,-5,6,-7,8]`

Table 10: Wyckoff bond: `1c@1b`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-2,-3,4,-5,6,7,-8]`

Table 11: Wyckoff bond: `2d@1b`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-2,-5,6]`
2	$[-X, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3,-4,-7,8]`

Table 12: Wyckoff bond: `2e@1b`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-3,-5,7]`
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[2,-4,-6,8]`

Table 13: Wyckoff bond: `2f@1b`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-4,-5,8]`
2	$[0, -Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[2,-3,-6,7]`

Table 14: Wyckoff bond: `4g@1b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-5]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[2,-6]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3,-7]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-8]`

* Wyckoff site: `1c`, site symmetry: `mmm`

Table 15: Wyckoff bond: `1a@1c`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,-3,-4,-5,-6,7,8]`

Table 16: Wyckoff bond: `1b@1c`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2,3,-4,-5,6,-7,8]`

Table 17: Wyckoff bond: `1c@1c`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2,-3,4,-5,6,7,-8]`

Table 18: Wyckoff bond: `2d@1c`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2,-5,6]`
2	$[-X, Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-4,-7,8]`

Table 19: Wyckoff bond: `2e@1c`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-3,-5,7]`
2	$[-X, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-4,-6,8]`

Table 20: Wyckoff bond: `2f@1c`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-4,-5,8]`
2	$[0, -Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-3,-6,7]`

Table 21: Wyckoff bond: `4g@1c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, \frac{1}{2}]$	`[1,-5]`
2	$[-X, -Y, Z]$	$[0, 0, \frac{1}{2}]$	`[2,-6]`
3	$[-X, Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[3,-7]`
4	$[X, -Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[4,-8]`

* Wyckoff site: `1d`, site symmetry: `mmm`

Table 22: Wyckoff bond: `1a@1d`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1,2,-3,-4,-5,-6,7,8]`

Table 23: Wyckoff bond: `1b@1d`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, Y, 0]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1,-2,3,-4,-5,6,-7,8]`

Table 24: Wyckoff bond: `1c@1d`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1,-2,-3,4,-5,6,7,-8]`

Table 25: Wyckoff bond: `2d@1d`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1,-2,-5,6]`
2	$[-X, Y, 0]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[3,-4,-7,8]`

Table 26: Wyckoff bond: `2e@1d`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1,-3,-5,7]`
2	$[-X, 0, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[2,-4,-6,8]`

Table 27: Wyckoff bond: `2f@1d`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1,-4,-5,8]`
2	$[0, -Y, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[2,-3,-6,7]`

Table 28: Wyckoff bond: `4g@1d`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[1,-5]`
2	$[-X, -Y, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[2,-6]`
3	$[-X, Y, -Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[3,-7]`
4	$[X, -Y, -Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[4,-8]`

* Wyckoff site: `1e`, site symmetry: `mmm`

Table 29: Wyckoff bond: `1a@1e`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, \frac{1}{2}, 0]$	`[1,2,-3,-4,-5,-6,7,8]`

Table 30: Wyckoff bond: `1b@1e`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, Y, 0]$	$[0, \frac{1}{2}, 0]$	`[1,-2,3,-4,-5,6,-7,8]`

Table 31: Wyckoff bond: `1c@1e`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, \frac{1}{2}, 0]$	`[1,-2,-3,4,-5,6,7,-8]`

Table 32: Wyckoff bond: `2d@1e`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, \frac{1}{2}, 0]$	`[1,-2,-5,6]`
2	$[-X, Y, 0]$	$[0, \frac{1}{2}, 0]$	`[3,-4,-7,8]`

Table 33: Wyckoff bond: `2e@1e`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, \frac{1}{2}, 0]$	`[1,-3,-5,7]`
2	$[-X, 0, Z]$	$[0, \frac{1}{2}, 0]$	`[2,-4,-6,8]`

Table 34: Wyckoff bond: `2f@1e`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[0, \frac{1}{2}, 0]$	`[1,-4,-5,8]`
2	$[0, -Y, Z]$	$[0, \frac{1}{2}, 0]$	`[2,-3,-6,7]`

Table 35: Wyckoff bond: `4g@1e`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-5]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-6]`
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[3,-7]`
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[4,-8]`

* Wyckoff site: `1f`, site symmetry: `mmm`

Table 36: Wyckoff bond: `1a@1f`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,-3,-4,-5,-6,7,8]`

Table 37: Wyckoff bond: `1b@1f`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-2,3,-4,-5,6,-7,8]`

Table 38: Wyckoff bond: `1c@1f`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-2,-3,4,-5,6,7,-8]`

Table 39: Wyckoff bond: `2d@1f`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-2,-5,6]`
2	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-4,-7,8]`

Table 40: Wyckoff bond: `2e@1f`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-3,-5,7]`
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-4,-6,8]`

Table 41: Wyckoff bond: `2f@1f`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-4,-5,8]`
2	$[0, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-3,-6,7]`

Table 42: Wyckoff bond: `4g@1f`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-5]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-6]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-7]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,-8]`

* Wyckoff site: `1g`, site symmetry: `mmm`

Table 43: Wyckoff bond: `1a@1g`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,-3,-4,-5,-6,7,8]`

Table 44: Wyckoff bond: `1b@1g`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,3,-4,-5,6,-7,8]`

Table 45: Wyckoff bond: `1c@1g`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,-3,4,-5,6,7,-8]`

Table 46: Wyckoff bond: `2d@1g`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,-5,6]`
2	$[-X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-4,-7,8]`

Table 47: Wyckoff bond: `2e@1g`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-3,-5,7]`
2	$[-X, 0, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-4,-6,8]`

Table 48: Wyckoff bond: `2f@1g`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-4,-5,8]`
2	$[0, -Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-3,-6,7]`

Table 49: Wyckoff bond: `4g@1g`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-5]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-6]`
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-7]`
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-8]`

* Wyckoff site: `1h`, site symmetry: `mmm`

Table 50: Wyckoff bond: `1a@1h`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,-3,-4,-5,-6,7,8]`

Table 51: Wyckoff bond: `1b@1h`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,3,-4,-5,6,-7,8]`

Table 52: Wyckoff bond: `1c@1h`, bond symmetry: `mmm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,-3,4,-5,6,7,-8]`

Table 53: Wyckoff bond: `2d@1h`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,-5,6]`
2	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-4,-7,8]`

Table 54: Wyckoff bond: `2e@1h`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-3,-5,7]`
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-4,-6,8]`

Table 55: Wyckoff bond: `2f@1h`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-4,-5,8]`
2	$[0, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-3,-6,7]`

Table 56: Wyckoff bond: `4g@1h`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1,-5]`
2	$[-X, -Y, Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[2,-6]`
3	$[-X, Y, -Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[3,-7]`
4	$[X, -Y, -Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[4,-8]`

* Wyckoff site: `2i`, site symmetry: `2mm`

Table 57: Wyckoff bond: `2a@2i`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[x, 0, 0]$	`[1,-4,-6,7]`
2	$[0, 0, Z]$	$[-x, 0, 0]$	`[2,-3,-5,8]`

Table 58: Wyckoff bond: `2b@2i`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[0, Y, 0]$	$[x, 0, 0]$	`[1,-4,6,-7]`
2	$[0, -Y, 0]$	$[-x, 0, 0]$	`[2,-3,5,-8]`

Table 59: Wyckoff bond: `2c@2i`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	`[1,4,6,7]`
2	$[-X, 0, 0]$	$[-x, 0, 0]$	`[2,3,5,8]`

Table 60: Wyckoff bond: `4d@2i`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[x, 0, 0]$	`[1,6]`
2	$[-X, -Y, 0]$	$[-x, 0, 0]$	`[2,5]`
3	$[-X, Y, 0]$	$[-x, 0, 0]$	`[3,8]`
4	$[X, -Y, 0]$	$[x, 0, 0]$	`[4,7]`

Table 61: Wyckoff bond: `4e@2i`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[x, 0, 0]$	`[1,7]`
2	$[-X, 0, Z]$	$[-x, 0, 0]$	`[2,8]`
3	$[-X, 0, -Z]$	$[-x, 0, 0]$	`[3,5]`
4	$[X, 0, -Z]$	$[x, 0, 0]$	`[4,6]`

Table 62: Wyckoff bond: `4f@2i`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[x, 0, 0]$	`[1,-4]`
2	$[0, -Y, Z]$	$[-x, 0, 0]$	`[2,-3]`
3	$[0, -Y, -Z]$	$[-x, 0, 0]$	`[5,-8]`
4	$[0, Y, -Z]$	$[x, 0, 0]$	`[6,-7]`

Table 63: Wyckoff bond: `8g@2i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, 0, 0]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, 0, 0]$	`[3]`
4	$[X, -Y, -Z]$	$[x, 0, 0]$	`[4]`
5	$[-X, -Y, -Z]$	$[-x, 0, 0]$	`[5]`
6	$[X, Y, -Z]$	$[x, 0, 0]$	`[6]`
7	$[X, -Y, Z]$	$[x, 0, 0]$	`[7]`
8	$[-X, Y, Z]$	$[-x, 0, 0]$	`[8]`

* Wyckoff site: `2j`, site symmetry: `2mm`

Table 64: Wyckoff bond: `2a@2j`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,-4,-6,7]`
2	$[0, 0, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2,-3,-5,8]`

Table 65: Wyckoff bond: `2b@2j`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,-4,6,-7]`
2	$[0, -Y, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2,-3,5,-8]`

Table 66: Wyckoff bond: `2c@2j`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,4,6,7]`
2	$[-X, 0, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2,3,5,8]`

Table 67: Wyckoff bond: `4d@2j`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,6]`
2	$[-X, -Y, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2,5]`

continued ...

Table 67

No.	vector	center	mapping
3	$[-X, Y, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	`[3,8]`
4	$[X, -Y, 0]$	$\left[x, 0, \frac{1}{2}\right]$	`[4,7]`

Table 68: Wyckoff bond: `4e@2j`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,7]`
2	$[-X, 0, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2,8]`
3	$[-X, 0, -Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[3,5]`
4	$[X, 0, -Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[4,6]`

Table 69: Wyckoff bond: `4f@2j`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,-4]`
2	$[0, -Y, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2,-3]`
3	$[0, -Y, -Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[5,-8]`
4	$[0, Y, -Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[6,-7]`

Table 70: Wyckoff bond: `8g@2j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[5]`
6	$[X, Y, -Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `2k`, site symmetry: `2mm`

Table 71: Wyckoff bond: `2a@2k`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[x, \frac{1}{2}, 0\right]$	`[1,-4,-6,7]`
2	$[0, 0, Z]$	$\left[-x, \frac{1}{2}, 0\right]$	`[2,-3,-5,8]`

Table 72: Wyckoff bond: `2b@2k`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[0,\ Y,\ 0]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,-4,6,-7]`
2	$[0,\ -Y,\ 0]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[2,-3,5,-8]`

Table 73: Wyckoff bond: `2c@2k`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,4,6,7]`
2	$[-X,\ 0,\ 0]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[2,3,5,8]`

Table 74: Wyckoff bond: `4d@2k`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,6]`
2	$[-X,\ -Y,\ 0]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[2,5]`
3	$[-X,\ Y,\ 0]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[3,8]`
4	$[X,\ -Y,\ 0]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[4,7]`

Table 75: Wyckoff bond: `4e@2k`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,7]`
2	$[-X,\ 0,\ Z]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[2,8]`
3	$[-X,\ 0,\ -Z]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[3,5]`
4	$[X,\ 0,\ -Z]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[4,6]`

Table 76: Wyckoff bond: `4f@2k`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ Y,\ Z]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,-4]`
2	$[0,\ -Y,\ Z]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[2,-3]`
3	$[0,\ -Y,\ -Z]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[5,-8]`
4	$[0,\ Y,\ -Z]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[6,-7]`

Table 77: Wyckoff bond: `8g@2k`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[4]`
5	$[-X,\ -Y,\ -Z]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[5]`

continued ...

Table 77

No.	vector	center	mapping
6	$[X, Y, -Z]$	$\left[x, \frac{1}{2}, 0\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[x, \frac{1}{2}, 0\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[-x, \frac{1}{2}, 0\right]$	`[8]`

* Wyckoff site: `2l`, site symmetry: `2mm`

Table 78: Wyckoff bond: `2a@2l`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-4,-6,7]`
2	$[0, 0, Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-3,-5,8]`

Table 79: Wyckoff bond: `2b@2l`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-4,6,-7]`
2	$[0, -Y, 0]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-3,5,-8]`

Table 80: Wyckoff bond: `2c@2l`, bond symmetry: `2mm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1,4,6,7]`
2	$[-X, 0, 0]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,5,8]`

Table 81: Wyckoff bond: `4d@2l`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1,6]`
2	$[-X, -Y, 0]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2,5]`
3	$[-X, Y, 0]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[3,8]`
4	$[X, -Y, 0]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[4,7]`

Table 82: Wyckoff bond: `4e@2l`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1,7]`
2	$[-X, 0, Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2,8]`
3	$[-X, 0, -Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[3,5]`
4	$[X, 0, -Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[4,6]`

Table 83: Wyckoff bond: `4f@2l`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-4]`
2	$[0, -Y, Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-3]`
3	$[0, -Y, -Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-8]`
4	$[0, Y, -Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[6,-7]`

Table 84: Wyckoff bond: `8g@2l`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[5]`
6	$[X, Y, -Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `2m`, site symmetry: `m2m`

Table 85: Wyckoff bond: `2a@2m`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, y, 0]$	`[1,-3,-6,8]`
2	$[0, 0, Z]$	$[0, -y, 0]$	`[2,-4,-5,7]`

Table 86: Wyckoff bond: `2b@2m`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$[0, y, 0]$	`[1,3,6,8]`
2	$[0, -Y, 0]$	$[0, -y, 0]$	`[2,4,5,7]`

Table 87: Wyckoff bond: `2c@2m`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, y, 0]$	`[1,-3,6,-8]`
2	$[-X, 0, 0]$	$[0, -y, 0]$	`[2,-4,5,-7]`

Table 88: Wyckoff bond: `4d@2m`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, y, 0]$	`[1,6]`
2	$[-X, -Y, 0]$	$[0, -y, 0]$	`[2,5]`

continued ...

Table 88

No.	vector	center	mapping
3	$[-X, Y, 0]$	$[0, y, 0]$	`[3,8]`
4	$[X, -Y, 0]$	$[0, -y, 0]$	`[4,7]`

Table 89: Wyckoff bond: `4e@2m`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, y, 0]$	`[1,-3]`
2	$[-X, 0, Z]$	$[0, -y, 0]$	`[2,-4]`
3	$[-X, 0, -Z]$	$[0, -y, 0]$	`[5,-7]`
4	$[X, 0, -Z]$	$[0, y, 0]$	`[6,-8]`

Table 90: Wyckoff bond: `4f@2m`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[0, y, 0]$	`[1,8]`
2	$[0, -Y, Z]$	$[0, -y, 0]$	`[2,7]`
3	$[0, Y, -Z]$	$[0, y, 0]$	`[3,6]`
4	$[0, -Y, -Z]$	$[0, -y, 0]$	`[4,5]`

Table 91: Wyckoff bond: `8g@2m`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[0, -y, 0]$	`[2]`
3	$[-X, Y, -Z]$	$[0, y, 0]$	`[3]`
4	$[X, -Y, -Z]$	$[0, -y, 0]$	`[4]`
5	$[-X, -Y, -Z]$	$[0, -y, 0]$	`[5]`
6	$[X, Y, -Z]$	$[0, y, 0]$	`[6]`
7	$[X, -Y, Z]$	$[0, -y, 0]$	`[7]`
8	$[-X, Y, Z]$	$[0, y, 0]$	`[8]`

* Wyckoff site: `2n`, site symmetry: `m2m`

Table 92: Wyckoff bond: `2a@2n`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, y, \frac{1}{2}\right]$	`[1,-3,-6,8]`
2	$[0, 0, Z]$	$\left[0, -y, \frac{1}{2}\right]$	`[2,-4,-5,7]`

Table 93: Wyckoff bond: `2b@2n`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[0,\ Y,\ 0]$	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1,3,6,8]`
2	$[0,\ -Y,\ 0]$	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[2,4,5,7]`

Table 94: Wyckoff bond: `2c@2n`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1,-3,6,-8]`
2	$[-X,\ 0,\ 0]$	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[2,-4,5,-7]`

Table 95: Wyckoff bond: `4d@2n`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1,6]`
2	$[-X,\ -Y,\ 0]$	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[2,5]`
3	$[-X,\ Y,\ 0]$	$\left[0,\ y,\ \frac{1}{2}\right]$	`[3,8]`
4	$[X,\ -Y,\ 0]$	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[4,7]`

Table 96: Wyckoff bond: `4e@2n`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1,-3]`
2	$[-X,\ 0,\ Z]$	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[2,-4]`
3	$[-X,\ 0,\ -Z]$	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[5,-7]`
4	$[X,\ 0,\ -Z]$	$\left[0,\ y,\ \frac{1}{2}\right]$	`[6,-8]`

Table 97: Wyckoff bond: `4f@2n`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0,\ Y,\ Z]$	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1,8]`
2	$[0,\ -Y,\ Z]$	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[2,7]`
3	$[0,\ Y,\ -Z]$	$\left[0,\ y,\ \frac{1}{2}\right]$	`[3,6]`
4	$[0,\ -Y,\ -Z]$	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[4,5]`

Table 98: Wyckoff bond: `8g@2n`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[0,\ y,\ \frac{1}{2}\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[4]`
5	$[-X,\ -Y,\ -Z]$	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[5]`

continued ...

Table 98

No.	vector	center	mapping
6	$[X, Y, -Z]$	$\left[0, y, \frac{1}{2}\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[0, -y, \frac{1}{2}\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[0, y, \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `2o`, site symmetry: `m2m`

Table 99: Wyckoff bond: `2a@2o`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[1,-3,-6,8]`
2	$[0, 0, Z]$	$\left[\frac{1}{2}, -y, 0\right]$	`[2,-4,-5,7]`

Table 100: Wyckoff bond: `2b@2o`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{2}, y, 0\right]$	`[1,3,6,8]`
2	$[0, -Y, 0]$	$\left[\frac{1}{2}, -y, 0\right]$	`[2,4,5,7]`

Table 101: Wyckoff bond: `2c@2o`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, y, 0\right]$	`[1,-3,6,-8]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2}, -y, 0\right]$	`[2,-4,5,-7]`

Table 102: Wyckoff bond: `4d@2o`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, y, 0\right]$	`[1,6]`
2	$[-X, -Y, 0]$	$\left[\frac{1}{2}, -y, 0\right]$	`[2,5]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, y, 0\right]$	`[3,8]`
4	$[X, -Y, 0]$	$\left[\frac{1}{2}, -y, 0\right]$	`[4,7]`

Table 103: Wyckoff bond: `4e@2o`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[1,-3]`
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, -y, 0\right]$	`[2,-4]`
3	$[-X, 0, -Z]$	$\left[\frac{1}{2}, -y, 0\right]$	`[5,-7]`
4	$[X, 0, -Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[6,-8]`

Table 104: Wyckoff bond: `4f@2o`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[1,8]`
2	$[0, -Y, Z]$	$\left[\frac{1}{2}, -y, 0\right]$	`[2,7]`
3	$[0, Y, -Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[3,6]`
4	$[0, -Y, -Z]$	$\left[\frac{1}{2}, -y, 0\right]$	`[4,5]`

Table 105: Wyckoff bond: `8g@2o`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, -y, 0\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, -y, 0\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, -y, 0\right]$	`[5]`
6	$[X, Y, -Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[\frac{1}{2}, -y, 0\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[8]`

* Wyckoff site: `2p`, site symmetry: `m2m`

Table 106: Wyckoff bond: `2a@2p`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[1,-3,-6,8]`
2	$[0, 0, Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[2,-4,-5,7]`

Table 107: Wyckoff bond: `2b@2p`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[1,3,6,8]`
2	$[0, -Y, 0]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[2,4,5,7]`

Table 108: Wyckoff bond: `2c@2p`, bond symmetry: `m2m`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[1,-3,6,-8]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[2,-4,5,-7]`

Table 109: Wyckoff bond: `4d@2p`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[1,6]`
2	$[-X, -Y, 0]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[2,5]`

continued ...

Table 109

No.	vector	center	mapping
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[3,8]`
4	$[X, -Y, 0]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[4,7]`

Table 110: Wyckoff bond: `4e@2p`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[1,-3]`
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[2,-4]`
3	$[-X, 0, -Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[5,-7]`
4	$[X, 0, -Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[6,-8]`

Table 111: Wyckoff bond: `4f@2p`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[1,8]`
2	$[0, -Y, Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[2,7]`
3	$[0, Y, -Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[3,6]`
4	$[0, -Y, -Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[4,5]`

Table 112: Wyckoff bond: `8g@2p`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[5]`
6	$[X, Y, -Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `2q`, site symmetry: `mm2`

Table 113: Wyckoff bond: `2a@2q`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, z]$	`[1,2,7,8]`
2	$[0, 0, -Z]$	$[0, 0, -z]$	`[3,4,5,6]`

Table 114: Wyckoff bond: `2b@2q`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[0, Y, 0]$	$[0, 0, z]$	`[1,-2,-7,8]`
2	$[0, Y, 0]$	$[0, 0, -z]$	`[3,-4,-5,6]`

Table 115: Wyckoff bond: `2c@2q`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, 0, z]$	`[1,-2,7,-8]`
2	$[-X, 0, 0]$	$[0, 0, -z]$	`[3,-4,5,-6]`

Table 116: Wyckoff bond: `4d@2q`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, z]$	`[1,-2]`
2	$[-X, Y, 0]$	$[0, 0, -z]$	`[3,-4]`
3	$[-X, -Y, 0]$	$[0, 0, -z]$	`[5,-6]`
4	$[X, -Y, 0]$	$[0, 0, z]$	`[7,-8]`

Table 117: Wyckoff bond: `4e@2q`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, z]$	`[1,7]`
2	$[-X, 0, Z]$	$[0, 0, z]$	`[2,8]`
3	$[-X, 0, -Z]$	$[0, 0, -z]$	`[3,5]`
4	$[X, 0, -Z]$	$[0, 0, -z]$	`[4,6]`

Table 118: Wyckoff bond: `4f@2q`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[0, 0, z]$	`[1,8]`
2	$[0, -Y, Z]$	$[0, 0, z]$	`[2,7]`
3	$[0, Y, -Z]$	$[0, 0, -z]$	`[3,6]`
4	$[0, -Y, -Z]$	$[0, 0, -z]$	`[4,5]`

Table 119: Wyckoff bond: `8g@2q`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, z]$	`[2]`
3	$[-X, Y, -Z]$	$[0, 0, -z]$	`[3]`
4	$[X, -Y, -Z]$	$[0, 0, -z]$	`[4]`
5	$[-X, -Y, -Z]$	$[0, 0, -z]$	`[5]`

continued ...

Table 119

No.	vector	center	mapping
6	$[X, Y, -Z]$	$[0, 0, -z]$	`[6]`
7	$[X, -Y, Z]$	$[0, 0, z]$	`[7]`
8	$[-X, Y, Z]$	$[0, 0, z]$	`[8]`

* Wyckoff site: `2r`, site symmetry: `mm2`

Table 120: Wyckoff bond: `2a@2r`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,2,7,8]`
2	$[0, 0, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[3,4,5,6]`

Table 121: Wyckoff bond: `2b@2r`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2,-7,8]`
2	$[0, Y, 0]$	$\left[0, \frac{1}{2}, -z\right]$	`[3,-4,-5,6]`

Table 122: Wyckoff bond: `2c@2r`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2,7,-8]`
2	$[-X, 0, 0]$	$\left[0, \frac{1}{2}, -z\right]$	`[3,-4,5,-6]`

Table 123: Wyckoff bond: `4d@2r`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2]`
2	$[-X, Y, 0]$	$\left[0, \frac{1}{2}, -z\right]$	`[3,-4]`
3	$[-X, -Y, 0]$	$\left[0, \frac{1}{2}, -z\right]$	`[5,-6]`
4	$[X, -Y, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[7,-8]`

Table 124: Wyckoff bond: `4e@2r`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,7]`
2	$[-X, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2,8]`
3	$[-X, 0, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[3,5]`
4	$[X, 0, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[4,6]`

Table 125: Wyckoff bond: `4f@2r`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0,\ Y,\ Z]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,8]`
2	$[0,\ -Y,\ Z]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[2,7]`
3	$[0,\ Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ -z\right]$	`[3,6]`
4	$[0,\ -Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ -z\right]$	`[4,5]`

Table 126: Wyckoff bond: `8g@2r`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ -z\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ -z\right]$	`[4]`
5	$[-X,\ -Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ -z\right]$	`[5]`
6	$[X,\ Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ -z\right]$	`[6]`
7	$[X,\ -Y,\ Z]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[7]`
8	$[-X,\ Y,\ Z]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[8]`

* Wyckoff site: `2s`, site symmetry: `mm2`

Table 127: Wyckoff bond: `2a@2s`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,2,7,8]`
2	$[0,\ 0,\ -Z]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[3,4,5,6]`

Table 128: Wyckoff bond: `2b@2s`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[0,\ Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,-2,-7,8]`
2	$[0,\ Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[3,-4,-5,6]`

Table 129: Wyckoff bond: `2c@2s`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,-2,7,-8]`
2	$[-X,\ 0,\ 0]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[3,-4,5,-6]`

Table 130: Wyckoff bond: `4d@2s`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,-2]`
2	$[-X,\ Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[3,-4]`

continued ...

Table 130

No.	vector	center	mapping
3	$[-X, -Y, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[5,-6]`
4	$[X, -Y, 0]$	$\left[\frac{1}{2}, 0, z\right]$	`[7,-8]`

Table 131: Wyckoff bond: `4e@2s`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[1,7]`
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[2,8]`
3	$[-X, 0, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[3,5]`
4	$[X, 0, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[4,6]`

Table 132: Wyckoff bond: `4f@2s`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[1,8]`
2	$[0, -Y, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[2,7]`
3	$[0, Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[3,6]`
4	$[0, -Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[4,5]`

Table 133: Wyckoff bond: `8g@2s`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[5]`
6	$[X, Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[8]`

* Wyckoff site: `2t`, site symmetry: `mm2`

Table 134: Wyckoff bond: `2a@2t`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,2,7,8]`
2	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[3,4,5,6]`

Table 135: Wyckoff bond: `2b@2t`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[0,\ Y,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,-2,-7,8]`
2	$[0,\ Y,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[3,-4,-5,6]`

Table 136: Wyckoff bond: `2c@2t`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,-2,7,-8]`
2	$[-X,\ 0,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[3,-4,5,-6]`

Table 137: Wyckoff bond: `4d@2t`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,-2]`
2	$[-X,\ Y,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[3,-4]`
3	$[-X,\ -Y,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[5,-6]`
4	$[X,\ -Y,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[7,-8]`

Table 138: Wyckoff bond: `4e@2t`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,7]`
2	$[-X,\ 0,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[2,8]`
3	$[-X,\ 0,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[3,5]`
4	$[X,\ 0,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[4,6]`

Table 139: Wyckoff bond: `4f@2t`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0,\ Y,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,8]`
2	$[0,\ -Y,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[2,7]`
3	$[0,\ Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[3,6]`
4	$[0,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[4,5]`

Table 140: Wyckoff bond: `8g@2t`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[4]`
5	$[-X,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[5]`

continued ...

Table 140

No.	vector	center	mapping
6	$[X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[8]`

* Wyckoff site: `4u`, site symmetry: `m..`

Table 141: Wyckoff bond: `4a@4u`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[0, y, z]$	`[1,8]`
2	$[0, -Y, Z]$	$[0, -y, z]$	`[2,7]`
3	$[0, Y, -Z]$	$[0, y, -z]$	`[3,6]`
4	$[0, -Y, -Z]$	$[0, -y, -z]$	`[4,5]`

Table 142: Wyckoff bond: `4b@4u`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, y, z]$	`[1,-8]`
2	$[-X, 0, 0]$	$[0, -y, z]$	`[2,-7]`
3	$[-X, 0, 0]$	$[0, y, -z]$	`[3,-6]`
4	$[X, 0, 0]$	$[0, -y, -z]$	`[4,-5]`

Table 143: Wyckoff bond: `8c@4u`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$[0, -y, z]$	`[2]`
3	$[-X, Y, -Z]$	$[0, y, -z]$	`[3]`
4	$[X, -Y, -Z]$	$[0, -y, -z]$	`[4]`
5	$[-X, -Y, -Z]$	$[0, -y, -z]$	`[5]`
6	$[X, Y, -Z]$	$[0, y, -z]$	`[6]`
7	$[X, -Y, Z]$	$[0, -y, z]$	`[7]`
8	$[-X, Y, Z]$	$[0, y, z]$	`[8]`

* Wyckoff site: `4v`, site symmetry: `m..`

Table 144: Wyckoff bond: `4a@4v`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[\frac{1}{2}, y, z\right]$	`[1,8]`
2	$[0, -Y, Z]$	$\left[\frac{1}{2}, -y, z\right]$	`[2,7]`
3	$[0, Y, -Z]$	$\left[\frac{1}{2}, y, -z\right]$	`[3,6]`
4	$[0, -Y, -Z]$	$\left[\frac{1}{2}, -y, -z\right]$	`[4,5]`

Table 145: Wyckoff bond: `4b@4v`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, y, z\right]$	`[1,-8]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2}, -y, z\right]$	`[2,-7]`
3	$[-X, 0, 0]$	$\left[\frac{1}{2}, y, -z\right]$	`[3,-6]`
4	$[X, 0, 0]$	$\left[\frac{1}{2}, -y, -z\right]$	`[4,-5]`

Table 146: Wyckoff bond: `8c@4v`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, y, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, -y, z\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, y, -z\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, -y, -z\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, -y, -z\right]$	`[5]`
6	$[X, Y, -Z]$	$\left[\frac{1}{2}, y, -z\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[\frac{1}{2}, -y, z\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[\frac{1}{2}, y, z\right]$	`[8]`

* Wyckoff site: `4w`, site symmetry: `.m.`

Table 147: Wyckoff bond: `4a@4w`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[x, 0, z]$	`[1,7]`
2	$[-X, 0, Z]$	$[-x, 0, z]$	`[2,8]`
3	$[-X, 0, -Z]$	$[-x, 0, -z]$	`[3,5]`
4	$[X, 0, -Z]$	$[x, 0, -z]$	`[4,6]`

Table 148: Wyckoff bond: `4b@4w`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[0, Y, 0]$	$[x, 0, z]$	`[1,-7]`
2	$[0, -Y, 0]$	$[-x, 0, z]$	`[2,-8]`
3	$[0, Y, 0]$	$[-x, 0, -z]$	`[3,-5]`
4	$[0, -Y, 0]$	$[x, 0, -z]$	`[4,-6]`

Table 149: Wyckoff bond: `8c@4w`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, 0, z]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, 0, -z]$	`[3]`
4	$[X, -Y, -Z]$	$[x, 0, -z]$	`[4]`
5	$[-X, -Y, -Z]$	$[-x, 0, -z]$	`[5]`

continued ...

Table 149

No.	vector	center	mapping
6	$[X, Y, -Z]$	$[x, 0, -z]$	[6]
7	$[X, -Y, Z]$	$[x, 0, z]$	[7]
8	$[-X, Y, Z]$	$[-x, 0, z]$	[8]

* Wyckoff site: 4x, site symmetry: .m.

Table 150: Wyckoff bond: 4a@4x, bond symmetry: .m.

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[x, \frac{1}{2}, z\right]$	[1,7]
2	$[-X, 0, Z]$	$\left[-x, \frac{1}{2}, z\right]$	[2,8]
3	$[-X, 0, -Z]$	$\left[-x, \frac{1}{2}, -z\right]$	[3,5]
4	$[X, 0, -Z]$	$\left[x, \frac{1}{2}, -z\right]$	[4,6]

Table 151: Wyckoff bond: 4b@4x, bond symmetry: .m.

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[x, \frac{1}{2}, z\right]$	[1,-7]
2	$[0, -Y, 0]$	$\left[-x, \frac{1}{2}, z\right]$	[2,-8]
3	$[0, Y, 0]$	$\left[-x, \frac{1}{2}, -z\right]$	[3,-5]
4	$[0, -Y, 0]$	$\left[x, \frac{1}{2}, -z\right]$	[4,-6]

Table 152: Wyckoff bond: 8c@4x, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{2}, z\right]$	[1]
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}, z\right]$	[2]
3	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}, -z\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}, -z\right]$	[4]
5	$[-X, -Y, -Z]$	$\left[-x, \frac{1}{2}, -z\right]$	[5]
6	$[X, Y, -Z]$	$\left[x, \frac{1}{2}, -z\right]$	[6]
7	$[X, -Y, Z]$	$\left[x, \frac{1}{2}, z\right]$	[7]
8	$[-X, Y, Z]$	$\left[-x, \frac{1}{2}, z\right]$	[8]

* Wyckoff site: 4y, site symmetry: ..m

Table 153: Wyckoff bond: 4a@4y, bond symmetry: ..m

No.	vector	center	mapping
1	$[X, Y, 0]$	$[x, y, 0]$	[1,6]
2	$[-X, -Y, 0]$	$[-x, -y, 0]$	[2,5]
3	$[-X, Y, 0]$	$[-x, y, 0]$	[3,8]
4	$[X, -Y, 0]$	$[x, -y, 0]$	[4,7]

Table 154: Wyckoff bond: `4b@4y`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[x, y, 0]$	`[1,-6]`
2	$[0, 0, Z]$	$[-x, -y, 0]$	`[2,-5]`
3	$[0, 0, -Z]$	$[-x, y, 0]$	`[3,-8]`
4	$[0, 0, -Z]$	$[x, -y, 0]$	`[4,-7]`

Table 155: Wyckoff bond: `8c@4y`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -y, 0]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, y, 0]$	`[3]`
4	$[X, -Y, -Z]$	$[x, -y, 0]$	`[4]`
5	$[-X, -Y, -Z]$	$[-x, -y, 0]$	`[5]`
6	$[X, Y, -Z]$	$[x, y, 0]$	`[6]`
7	$[X, -Y, Z]$	$[x, -y, 0]$	`[7]`
8	$[-X, Y, Z]$	$[-x, y, 0]$	`[8]`

* Wyckoff site: `4z`, site symmetry: `..m`

Table 156: Wyckoff bond: `4a@4z`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[x, y, \frac{1}{2}\right]$	`[1,6]`
2	$[-X, -Y, 0]$	$\left[-x, -y, \frac{1}{2}\right]$	`[2,5]`
3	$[-X, Y, 0]$	$\left[-x, y, \frac{1}{2}\right]$	`[3,8]`
4	$[X, -Y, 0]$	$\left[x, -y, \frac{1}{2}\right]$	`[4,7]`

Table 157: Wyckoff bond: `4b@4z`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[x, y, \frac{1}{2}\right]$	`[1,-6]`
2	$[0, 0, Z]$	$\left[-x, -y, \frac{1}{2}\right]$	`[2,-5]`
3	$[0, 0, -Z]$	$\left[-x, y, \frac{1}{2}\right]$	`[3,-8]`
4	$[0, 0, -Z]$	$\left[x, -y, \frac{1}{2}\right]$	`[4,-7]`

Table 158: Wyckoff bond: `8c@4z`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, y, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, -y, \frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[-x, y, \frac{1}{2}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x, -y, \frac{1}{2}\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[-x, -y, \frac{1}{2}\right]$	`[5]`

continued ...

Table 158

No.	vector	center	mapping
6	$[X, Y, -Z]$	$\left[x, y, \frac{1}{2}\right]$	[6]
7	$[X, -Y, Z]$	$\left[x, -y, \frac{1}{2}\right]$	[7]
8	$[-X, Y, Z]$	$\left[-x, y, \frac{1}{2}\right]$	[8]

* Wyckoff site: 8A, site symmetry: 1

Table 159: Wyckoff bond: 8a@8A, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	[1]
2	$[-X, -Y, Z]$	$[-x, -y, z]$	[2]
3	$[-X, Y, -Z]$	$[-x, y, -z]$	[3]
4	$[X, -Y, -Z]$	$[x, -y, -z]$	[4]
5	$[-X, -Y, -Z]$	$[-x, -y, -z]$	[5]
6	$[X, Y, -Z]$	$[x, y, -z]$	[6]
7	$[X, -Y, Z]$	$[x, -y, z]$	[7]
8	$[-X, Y, Z]$	$[-x, y, z]$	[8]